Two-Seller Competition with Partial Information

Progress note — transcription of handwritten notes

October 11, 2026

This note transcribes the model in the handwritten notes. We use the coordinate order (x_A, x_B) throughout and write seller-specific expressions with a consistent index $j \in \{A, B\}$. Questions in the notes about the boundary and the stationary distribution remain open.

1 Environment, information, and timing

Consider two sellers, indexed by $j \in \{A, B\}$, each with two products, $i \in \{1, 2\}$. Product i generates a successful outcome with probability p_i. The model incorporates forgetting with parameter $\rho \in (0, 1)$. The state at time t is

$$\mathbf{x}_t = (x_{A,t}, x_{B,t}), \tag{1}$$

where $x_{j,t}$ summarizes seller j's successes under forgetting. Each seller observes only its own outcomes and state $x_{j,t}$, not the other seller's state. Write $-j$ for seller j's opponent.

Following the restriction in the notes, we work with

$$\rho \in \left(\frac{1}{2}, 1\right), \qquad K_\rho = [0, 1], \qquad \mathbf{x}_t \in K_\rho^2. \tag{2}$$

This restriction is intended to avoid the Cantor-set issue associated with smaller values of ρ. The notes ask whether the proposed state representation amounts to working on the boundary of the original forgetting model; this interpretation remains to be clarified.

Timing. Let $y_t \in \{A, B\}$ denote the customer's choice, $i_t \in \{1, 2\}$ the selected seller's product, and $u_t \in \{0, 1\}$ the realized outcome. Within each period:

1. The customer chooses a seller j.
2. The selected seller chooses a product i.
3. The outcome is realized, with success probability p_i, and the selected seller's state is updated.

In particular,

$$\mathbb{P}(u_t = 1 \mid y_t = j, i_t = i, \mathbf{x}_t) = p_i. \tag{3}$$

Customer choice. The selection probabilities are

$$\mathbb{P}(y_t = j \mid \mathbf{x}_t) = q_j(\mathbf{x}_t) = \Lambda(x_{j,t} - x_{-j,t}), \tag{4}$$

where $\Lambda : [-1, 1] \to [0, 1]$ is continuous and increasing, with

$$\Lambda(-z) = 1 - \Lambda(z). \tag{5}$$

Consequently, $q_A(\mathbf{x}) + q_B(\mathbf{x}) = 1$.

2 Strategies and the joint transition kernel

State updates. For a selected seller with current state x, define

$$T^s(x) = \rho x + (1-\rho), \qquad T^f(x) = \rho x, \tag{6}$$

where s and f denote success and failure. The unselected seller's state remains unchanged.

Strategies. Fix a strategy profile $\sigma = (\sigma_A, \sigma_B)$, where

$$\sigma_j : K_\rho \longrightarrow \Delta(\{1,2\}). \tag{7}$$

Here $\Delta(\{1,2\})$ is the set of probability distributions over the two products, and $\sigma_j(i \mid x_j)$ is the probability that seller j chooses product i when selected at its own state x_j. The resulting average success probability is

$$\bar{p}_j(x_j) = \sum_{i=1}^{2} \sigma_j(i \mid x_j) p_i. \tag{8}$$

From a joint state $\mathbf{x} = (x_A, x_B)$, the four possible events have the following next states and probabilities:

Event	Next state	Probability
A selected, success	$(T^s(x_A), x_B)$	$q_A(\mathbf{x})\bar{p}_A(x_A)$
A selected, failure	$(T^f(x_A), x_B)$	$q_A(\mathbf{x})(1-\bar{p}_A(x_A))$
B selected, success	$(x_A, T^s(x_B))$	$(1-q_A(\mathbf{x}))\bar{p}_B(x_B)$
B selected, failure	$(x_A, T^f(x_B))$	$(1-q_A(\mathbf{x}))(1-\bar{p}_B(x_B))$

Let δ_z denote the Dirac probability measure at z; for a measurable set E, $\delta_z(E) = \mathbf{1}\{z \in E\}$. The joint transition kernel, the continuous-state counterpart of a transition matrix, is

$$\begin{aligned} P^\sigma(\mathbf{x}, \cdot) = q_A(\mathbf{x}) \Big[\bar{p}_A(x_A)\delta_{(T^s(x_A), x_B)} + (1-\bar{p}_A(x_A))\delta_{(T^f(x_A), x_B)} \Big] \\ + (1-q_A(\mathbf{x})) \Big[\bar{p}_B(x_B)\delta_{(x_A, T^s(x_B))} + (1-\bar{p}_B(x_B))\delta_{(x_A, T^f(x_B))} \Big]. \end{aligned} \tag{9}$$

3 Stationary distribution and expected demand

Let π^σ be an invariant probability distribution for P^σ, satisfying

$$\pi^\sigma(E) = \int_{K_\rho^2} P^\sigma(\mathbf{x}, E)\, \pi^\sigma(d\mathbf{x}) \qquad \text{for every measurable } E \subseteq K_\rho^2. \tag{10}$$

As in the notes, we assume that this distribution exists and is unique. The conditions that justify existence and uniqueness remain an open question.

For seller j, define stationary expected demand conditional on its own state by

$$\begin{aligned} \eta_j^\sigma(x) &:= \mathbb{E}^{\pi^\sigma}[q_j(\mathbf{x}_t) \mid x_{j,t} = x] \\ &= \int_{K_\rho} \Lambda(x - x')\, \pi^\sigma(dx' \mid x_j = x). \end{aligned} \tag{11}$$

Here $\pi^\sigma(dx' \mid x_j = x)$ denotes the conditional distribution of the opponent's state $x_{-j} = x'$ under π^σ. Thus $\eta_j^\sigma(x)$ depends only on seller j's own state: the opponent's unobserved state is averaged out using its conditional stationary distribution.

4 Transition law conditional on a seller's own state

Under this stationary demand, the one-period conditional distribution of seller j's next state is

$$\begin{aligned}&\mathbb{P}^{\pi^\sigma}(x_{j,t+1} \in \cdot \mid x_{j,t} = x)\\ &\quad = \eta_j^\sigma(x)\Big[\bar{p}_j(x)\delta_{T^s(x)} + (1-\bar{p}_j(x))\delta_{T^f(x)}\Big] + (1-\eta_j^\sigma(x))\delta_x.\end{aligned} \tag{12}$$

With probability $\eta_j^\sigma(x)$, seller j is selected and its state changes according to the outcome. Otherwise, its state remains at x. Every term in this expression depends on the seller's own state and the fixed strategy profile.

Technical clarification (not in the handwritten notes). Equation (12) is a one-period conditional law under the stationary joint distribution. By itself, it does not establish that a seller's own-state process is Markov with respect to its entire observed history. Using it as a complete dynamic model based only on x would require a separate justification.